

ActInf GuestStream #037.1 ~ Baba Brinkman

GuestStream | Baba Brinkman | 1:20:46

hello and welcome it's February 28 2023
it's active guest stream number 37.1 and
we're here with Baba Brinkman this
should be a great discussion an
emergence Baba thank you so much for
joining let's just jump right into it
what brings you to this day how and who
are you
and welcome
thanks thanks Daniel so much for having
me here it's a real uh it's a real honor
to join you and to uh you know dip into
this community which I've been aware of
for a while and uh you know like to try
to follow the work as much as I can but
I'm not a scientist myself uh I'm a
rapper I have been uh exploring academic
topics in hip-hop for quite a while I
did my Master's thesis on Chaucer's
Canterbury Tales I was a Renaissance and
medieval scholar uh and I I rewrote the
Canterbury Tales into rap and I used to
tour around the UK wrapping My Wife of
Bath's tail and Miller's tale Renditions
which was a lot of fun and then at one
point it was around 2007 or 2008 uh an
evolutionary biologist a geneticist saw
my Canterbury Tales show and he reached
out to me and said I love what you're
doing for medieval literature
popularizing and popularizing it with
rap but there's other topics that need
rap popularization I think his exact
words were would you be willing to do
for Darwin what you did for Chaucer and

this was a in the run-up to the uh the Darwin Bicentennial in 2009 it was the 200th anniversary of uh the birth of Charles Darwin also the 150th anniversary of the publication of the Origin of Species I was of course a big fan of Darwin and of evolutionary theory but not very expert so I said to this scientist yes I'll write some raps about Evolution but I could use some guidance uh he sent me a whole bunch of suggested reading sources books uh peer-reviewed papers everything so I took a sort of six month Deep dive on the state of evolutionary theory uh absorbed a whole bunch of books and uh and wrote These Rhymes but his uh his his criteria was your performances will be at academic seminars with experts in darwinian evolution across all kinds of social domains and psychology and everything so you need to make sure you represent the theory accurately which means you have to run your lyrics by me so I can check them for scientific veracity uh which I thought was a fair criteria for that kind of context so I sent him my lyrics he gave me some uh you know some revision suggestions and then when I got up when I was like set to perform the premiere of my Darwin show in February uh on Darwin day February 12 2009 uh the scientist his name is Mark Palin he was a bacterial geneticist he had this great intro typical British tongue-in-cheek style he was like I have fact check the lyrics and you're about to witness the first rap show in history that's peer

reviewed now like peer-reviewed rap
that's a great idea uh totally totally
marks coinage uh but then right away the
wheel started spinning like what other
topics could you apply the peer-reviewed
uh wrap lens to so I did my Evolution
rap show for several years it was called
the rap guide to Evolution it played Off
Broadway won an award at the Edinburgh
Fringe Festival and then other
scientists started tugging me on the
sleeve like oh the Darwin wraps are cool
but could you do
um Wilderness conservation and
threatened species ecology could you do
climate change could you do Neuroscience
uh and and various uh great
collaborations came from that so I now
have a whole series of rap albums rap
guides to various topics and my
encounter with the active inference uh
community and research base was when I
was working on one called the rap guide
to Consciousness and uh Professor Anil
Seth is one of the uh I think main
proponent opponents and contributors to
the Bayesian brain active inference
research Paradigm today and he was one
of the main science advisors on the
project so he was giving me Bayesian
stuff to read including Andy Clark's
what's next highly technical uh review
of all the literature which I slogged my
way through as a non-expert and ended up
working Bayesian ideas quite in a
detailed way into the rap guide to
Consciousness show and what really
resonated for me honestly I had never

really paid attention to Bayes theorem
or or Bayesian reasoning at all uh but I
you know it took me a while to get my
head around it but then I was like okay
this is actually just a great uh
paraphrase of the scientific method and
I was already a massive proponent of the
scientific method and I was geeking out
on the degree to which the brain may be
a Bayesian engine at the microscopic
level producing a rational inquiry
process that replicates the exact same
Bayesian algorithm at the level of
interactions with Society so I did one
song in the show called uh good Bayesian
let me show you how to be a good basy
and change your predictions after taking
information in and if you're thinking
I'll be less than amazing let's adjust
those expectations and I was right away
thinking about like how how is a rap
performance itself an instantiation of a
Bayesian algorithm with the audience's
expectations being revised in real time
based on the rapper's performance and
the rapper's expectations when creating
the script of the performance about what
reaction they'll get from the audience
which is a prior belief about the
effectiveness of the performance but
when it's only a script don't actually
know what's going to work and then you
do the performance you see what gets the
reaction you revise your priors and you
also revise your script and that was a
concept that I had already been rapping
about in my Evolution show I have a show
a whole song and that called performance

feedback revision which instantiates the kind of evolutionary natural selection algorithm but at the level of creative writing and performance you write the script you do the show you see how people react you go back you change the script and that's kind of like the genome being revised under phenotypic variants and how they end up Surviving or not surviving reproducing or not reproducing so

um you know this is I I just started out rapping and trying to master the craft and each scientific topic I get into I start going oh this actually helps me understand what's going on in rap much better and especially actually one of the examples I play with in the show which is a lot of fun is like uh call-in response expectations of the audience like a perfect kind of demonstration creation of a Bayesian moment when you go

you know Double Take look twice calculate the odds roll the and when I do that the whole crowd would be like dice you know but the expectation of what the rhyme couplet's gonna be is a game rappers play all the time like finish the finish the bar right and that is entirely around a setup of an expectation and the Fulfillment of it but then sometimes you can play with it and be like yo perceptual Basics ask Morpheus we live in the and I usually get the crowd being like Matrix back of me but then other times I'd make it more difficult like yo perceptual basics

ladybugs like to eat and then like maybe
one person out of 100 would be like
aphids and I'd be like yeah you know
that that was the person that had the
correct scientific Bayesian prior is to
conclude my Matrix Basics aphids rhyme
pattern but usually only a few get it
and then everyone else is like ah
there's the nerd over there or whatever
so you know it's it's become like
actually informed my creative process in
terms of how I try to set up and violate
expectations in real time
psychologically with an audience when
I'm performing
um and I will be honest that my close
reading of the research was several
years ago at this point uh the rap guide
to Consciousness came out in 2017 uh 18
I was performing it Off Broadway 19 I
was uh touring it around uh theaters in
the UK and still performing Off-Broadway
and then 20 of course all performances
ceased all together and I had to think
about what rappers do in pandemics when
they can't get on stages and see crowds
so uh yeah we can talk uh during this
interview about my new project which is
a custom wrap agency you can see me
wrapping it right here event wrap uh
which could be fun to talk about in
terms of the event inference uh elements
of active inference and and Bayesian
analysis could be uh you know I think
some of the aspects of what attract me
to events What attracted me to events
during the pandemics and still does as a
sort of a catalyst for good art and for

rap might have to do with active
inference uh expectations and
experiences but I don't know this
literature very closely at this point so
um yeah that's basically brings us up to
the present hopefully that was a good
summary
amazing uh just great to
connect and to be able to share this
moment with someone who's seen it from
all sides
in front of the stage on stage behind
stage Inside the Mind real recognize
real
so there's a lot out there and just like
as if we were talking about a paper
namaste real recognize real namaste
if we were talking about a paper
we would read between the lines and
reference out and similarly there's so
much that can be referenced out and
people can listen to the original
creations and see what that primary text
looks like and the fact that it's uh
orally performed text for much of human
development in history was our real
so whether something is written
is increasingly fluid these days with
speech to text and text-to-speech so
there's going to be so many great
Avenues let's just start with
event-based cognition so what is an
event from the cognitive perspective
I mean I was trying to find that
definition with Google about 20 minutes
ago so I didn't sound like an idiot in
this podcast so uh I think it's a uh you
know a

it's a perceptual construct right it is
the perception of an entity that there
is some kind of instantiation of action
that is circumscribed in time that has a
beginning and an end and if it's not
perceived as having a beginning and an
end by some cognitive entity
historically organisms but in the future
also probably AIS uh then it's not you
know then it's not treated as an event
which means an event could be a thousand
years or five seconds or whatever it's
it's actually interesting how the kind
of the scientific definition of it in
active inference research really is like
a narrative definition in a way right
it's like it's like it's a story
something you can tell a story about
something you can construct a story
around uh is an event is that pretty
close to how scientists think about it
yeah I mean already so much there you're
right it does not have to be tied to a
given amount of clock time which we
might call chronology hearkening back to
Kronos that sort of decimal point time
but action takes place
in the understanding of the actor or the
observer in this time scale of action or
Kairos and that time scale of action
may be narrative and that is what you
invoke with the beginning middle and end
rising and falling cultural patterns
that's not necessarily happening on the
clock just because you can run a
stopwatch while somebody's reciting
Canterbury Tales
doesn't mean that

it'd be twice as exciting if they spoke
twice as fast or that you could
map in some linear way what was
happening on the watch face to what's
happening in the mind of audience who
have different familiarity with the
material or a historical study context
as well it just made me think of like if
I do a 10 minute long wrap of The Wife
of Bath's Tale then that's an event I
performed the wife of bathdale there was
an audience that witnessed it you could
circumscribe it that way but you could
also you know so you could do a
scientific study on what occurred during
that event right but you could also
choose to study the event of humanities
retelling of the story of The Wife of
Bath's tail which is a story that takes
place in King Arthur's Court so now we
know where the mythology has to have
started which is around you know forgive
me medievalists but let's say like four
or 500 A.D and then whenever the
historical King Arthur's Court existed
and then there was an oral mythology
that adhered to that over the course of
a period of time as people told folk
tales about it and then in the year 1400
uh Jeffrey Chaucer uh
well I think he perished in 1400 so he
was born in the 1330s but he writes the
wife of bath tale at some point as a
written representation of the oral
narrative and then that Vellum
manuscript is passed down into college
reading lists and I read it in 1998 and
write a rap version of it so any study

of Humanity's engagement with the story of the wife of Bastille has to really start uh you know in the year 400 and and the last time I wrapped my rap version of the wife of Bastille at the Edinburgh Fringe Festival which was a couple years ago and no event that accounts for the phenomenon of the telling of that story can leave any of those instantiations out right awesome it's like the earlier events are our prior or our generative process which is our Niche that events happen in our generative model our cognitive model exists within the niche the generative process and those prior events quite literally are our priors and there's no last chapter because the observations keep rolling in and whether the observation is exactly as expected whether that performance of that symphony is exactly as expected but they're that performance of the hamburger or of the portrait even if it's exactly as expected it still is a new observation and then what is as expected well that's in the encultured cognitive mind and that is this sometimes implicit reference point that novelty is contrasted against like well I knew I was going to get a rap summary but I didn't know it'd be that funny or I didn't know I was going to get a rap summary these are like different levels and types of novelty about the story of a moment or or look at how he changed the story from the archival version

because there's definitely a completely different experience of people that are intimately familiar with trousers version and ones who are hearing the story for the first time and encountering each of the story the characters this is the kind of expectation of how accurately will be the story will the story be recounted in this new version uh also let me just um expand on something that you said he said there's no last prior because there's constant reinstantiations and updating right but there is a last prior from the perspective of a temporary analysis of an event you can sort of you can set a sort of artificial beginning and end for the purpose of doing an analysis and that's I guess where you're able to apply mathematics to it but you always have to like you know keep in mind that it's a chosen beginning and endpoint within a larger unfolding yeah and that's very um resonant with my experience as a researcher which is like there is a final data point in any time series at some point the fmri machine or the ant foraging watching machine you just turn it off for the day so there is empirically something that's very addressable but in the general case it's not the case that a Time series algorithm says what the last observation is and we might even say that that's a deflationary perspective on history because rather than debating whether

somebody has released the definitive American novel or the last word on this Symphony it's like well the last note of their Symphony was non-controversially the last another Symphony but the question about the discourse in the community

of course there's no last word in this open-ended space and art is always like surfing on that open space

one of the things I remember from my uh comparative literature courses is this idea of the anxiety of influence and how poets have their uh I think it's maybe Harold Bloom that wrote that but it you know poet poets have a sense of needing to create something new I think it's General like applies to Art all artists right theater rappers you know you know this is like imagination conjures but of course great art is produced on the shoulders of a lot of influence in the past and one of the measures in an English literature course of whether a poem Rises to the level of worthiness to be studied at University is the degree of impact it had afterwards and that can sometimes be measured by how many imitators it generated so one thing I always find fascinating is you read 14th century literature and Chaucer stands out as the main poet right if you barely read any 15th century literature in an English Lit course or at least when I was in school in the 90s that was the case I've just gone all blurry haven't I I'm gonna see what's going on with my camera here

um there we go refocused the camera has its own priors

um so uh you know like Chaucer shut down English literature for like a hundred years and when I was when I was in an English lyric survey course I'm like why do we read Chaucer in the 14th century and the next Port we read is the 16th century what happened to the 15th century and my professors were like well every poet that wrote poetry in the 15th century was seen as so much of a Chaucer knockoff and derivative because his style was so influential that none of them rise to the level of having contributed anything distinctive until over 100 years later I was like wow now that is impact right that is like as much impact as you but ironically it's like negative impact like the most significant poem shuts everybody up because they don't dare to do anything except for imitated in a totally derivative way but creating something new out of it you know that's what is creates the most fecundity in art right like you create something and a lot of her people were like oh I love this I'm going to take it and adapt it run with it that's one of the things I love about hip-hop is it's so much about sampling and flipping and remixing and it's not afraid of any other genre because its attitude is you can take any genre and make it better by adding beats and Rhymes uh but but really effective art shuts it down I think Andrew Marvel was also known as like the last Carpe Diem

poem writer because he wrote Such a perfect Carpe Diem poem Gather ye rosebuds while you may that no one really bothered with that genre anymore because nobody could do as well as Andrew Marvel and he wrote the last great Garden poem as well English garden poem and it's like you know how do you measure your success as an artist by the degree to which you've created something that's so intimidating for everybody else that they don't even bother with the genre anymore and now you're measuring your success by the deletion of genres from the cultural milieu which creates space for other people to invent more genres but this is an interesting way to measure success

wow

I'm gonna pick up on that pattern and bring it into attention and the Bayesian perspective on attention which is sometimes equated with precision so if we're paying a lot of attention to a sensory input a lot of attention High Precision we're really putting a lot of credence in that information it's overriding our priors a lot we're updating our beliefs to basically be what we're paying attention to

and then the other case would be we're not paying attention we have low Precision or high uncertainty and we're not updating our beliefs at all that's what we basically conversationally mean when we say we're not paying attention to that like I couldn't tell you what it

was it didn't leave an impact
and when you were talking about this
cultural evolutionary process I thought
about cultures paying attention to
genres and works because if there was
one work that was the object of
attention

it would be in the spotlight but
attention is like a spotlight and if
it's a laser on one thing it's not on
everything else

and if there's no work that's rising to
attention

then there's this proliferation of
hypotheses cultural endeavors
but no one of them will necessarily be
shared and so art that resonates with
communities and with your friends it's
like in this gray Zone where it has to
have enough attention on it so that
there can be a culture
of hip-hop or the Grateful Dead or
whatever it may be

but

not one that is so attended to
that it

gets burnt by the laser

yeah it's like if everyone's only paying
attention to one thing then that's
actually very stifling for cultural
flourishing because it's you know you
don't you don't get all the variance you
get these like totally dominant moments
you can see that at the macro level in
history but it also just happens in
social media all the time too right
everybody's talking about one thing and
then it's like

you know what what if that thing is not
interesting to you or you know
what that what is the upshot of that com
of that dominance of attention
in active inference we sometimes talk
about regimes of attention and there
have been some great recent papers on
cultural regimes of attention and that
kind of takes that basal Bayesian idea
of attention as Precision to stimuli
and it puts it into the dynamical space
where the regime of attention might be
like the sequence of eye fixations across
a page so that is a complex pattern with
attention moving and we could just call
that broadly like a regime of attention
or attention and action
and
in some sense that's what culture is or
can be described as as regime just the
kind of summary of where you happen to
look or is the intent that it's like
there is some kind of intentional
control of attention that is inherent to
the work that's being attended to like
you can expect people to look others in
the eyes when you're seeing a video or a
photo in the regime is that like there
must be some implied like uh maybe in
inbuilt psychology of things that we
tend to find more fascinating is that is
that what's implied by it or
a whole lot there the word regime does
have a slight connotation of something
that's imposed
whereas it is in the neutral sense
describing merely the Dynamics and then
what those shared regimes of attention

might be

like we're going to approach each other
this way making this kind of eye contact
and make these kinds of gestures that's
how we're going to be reducing our
uncertainty about who we are and what
this event is

and without those shibboleths
secret words and implicit or explicit
processes and rituals it's hard to lock
on culturally

and

there's so many ways to go from there

but I'll leave it yeah I'm always

whenever I see shibboleths and Norms being
acted out I'm always trying to I mean

obviously there's scientific studies

that are trying to unpack this as well

but my favorite thing to speculate about

is to what degree is this culturally

constructed in an open-ended way and to

what degree is the cultural aspects of

it uh relying on an underlying cognitive

psychology that is inherited From

Evolution like for instance you know if

you're having a conversation with

somebody one-on-one and you never look

them in the eyes

you know you that's inherently

suspicious right like what's wrong with

this person they won't look me in the

eyes are they lying to me do they have

something to hide like that response

that you get emotionally and in terms of

like somebody's disposition towards you

you I would find that very unlikely to

be 100 culturally constructed like could

you imagine a society that was

egalitarian where they never looked each other in the eyes I can imagine societies where Aristocrats prohibit uh people below them in the hierarchy from looking in the eyes because it's seen as a sign of disrespect which of course is entirely based on primate behavior when you're like challenging an alpha if you look at them for too long then they'll bite you and you gotta like cast your eyes down to demonstrate your submissiveness or whatever like humans have replicated some of these very deep mammalian instincts that we have right but like are we malleable enough that a society could ever evolve with the norm of nobody ever looks the other person in the eyes or would that stream swim upstream so strongly against our instinctive ways of measuring our degrees of trust and Affinity towards each other that that culture would just inherently fail and and not manage to be represented in the panoply of cultures going forward

I'll try to give a thought on that falling back to this attention discussion so the movement of the eyes does provide information whether it's deceptive or not it is an outlet of information from the person who you could be looking at so it's putting attention itself right where you look to find out if someone's paying attention

the eyes are the window to the soul maybe that wasn't just about the cornea attention as the window to the soul

it was about Game Theory
and so to neglect this Fountain of
information
you have to have a cultural clamp on it
like a secondary reason to not look
because it is providing so much info on
a millisecond by millisecond level in
ways that aren't always even expressible
like pupil diameter has a lot to do with
arousal and psychologically people may
or may not know about that empirically
and I doubt it was shown empirically
until we had cameras and measuring
devices and millimeters and all of that
but it can still play a role implicitly
in behaviors and heuristics and
strategies and so it's like if we're
paying attention to our embodiment
certain things will arise
and
in situations where the inherited prior
has to overrule or reinterpret those
first principles takes
those are like scenarios with an
attention to the Past
for better and worse
that is shading
our experience interpretation
versus the Prima fascia situational
awareness that convergently evolves to a
few attractors like looking at
informative sources of information
yeah and and just because there's a lot
of uh swimming Upstream of uh sort of
instinctive cognition necessary for uh
cultural norms to play out doesn't
necessarily mean that they're doomed to
fail or you know a waste of time or

whatever you could almost say like the whole scientific Enterprise complex mathematics even reading none of these things are you know instinctively evolved they're all very recent and we're like building them on top of our uh older cognition and and architecture and instincts but they're you know gloriously culturally constructed and I would say like worthy of uh the effort of swimming Upstream

the keyboard

q-w-e-r-t-y oh well the keyboard doesn't have to be that way well yeah but we don't need keyboards and we haven't always had them and we won't always have them and left to right reading it could be right to left or up to down or it could be any but we haven't even had inscription technology

so maybe let me bring it back to your experience with hip-hop how has the generative process the extended cognitive niche of Technology supported the remix culture and how does that manifest

well I think I'll start with some of the good ones I used to go to a studio and sit with producers and hear their beats and uh you know like let's go with that one and then like Try lyric side pre-written or sometimes write the track um it was really expensive

fun interactively I I do miss the in-person studio experience but all the previous rap records I've made for the past 10 years were all made remotely with Dropbox so the rappers you know

send or the producers send me a folder full of Beats I choose the one that I like the best I write the rap I record the wraps at home I send them to the producer and I've got like a track where the producers based in Seattle the uh session musician is playing Keys is in Vancouver the vocalist is doing the song uh you know from his place in Toronto and I'm in Long Island like taking all the different parts and editing them and then sending them to the mix engineer and the master you know it's like this great distributed collaborative process and you know that's even by today's standard that's analog right that's like sharing files in a Dropbox and throwing them into your local studio and and sort of like measuring each input against the expectation a song to sound ultimately and giving people feedback and asking for revisions uh but I think there are now ai uh sort of web 3 platforms where you can all be in the same virtual room and the latencies worked out in your collaborating musically I haven't used those yet but I've been uh you know I've been scoping them out so that's been a very I think that's generally been positive it's brought the cost down and it's allowed for real time well not quite real time but like ongoing collaboration with creatives across different areas uh but then the kind of parallel Revolution that's happened is from selling CDs uh selling tape selling

CDs and then Spotify takes over and you're getting like fractions of pennies on the stream and your music is going into this massive Archive of every kind of music and you're only getting paid based on how much your music is listened to compared to every other music that's on there they're partitioning their subscription fees and it's really it's become much harder uh I think for a lot of musicians to make a living from the sale of music and so over the past 10 to 15 years that process really pushed a lot of artists into really just basing all their income on live performance like setup gigs and and Rock shows and sell Merchant shows if you want any expectation of income from music sales you better be there meeting the fan shaking their hand and sending them a CD in person then kovid shut that down you know for a couple of years anyway it's coming back but my estimation during the pandemic was even as it comes back it's gonna you know there's gonna be people that are more reluctant to travel to shows and you know do that have the in-person experience even when Public Health notices say it's safe because people have different degrees of paranoia and anxiety and they won't feel as comfortable even if it is objectively safe uh so you know pivoting to this uh event rap game was partially based on an analysis of what's happening with the industry and how hard it is to make it as an artist and part of that has been uh you know a technology driven

process

wow also many ways to go

let's kind of return to those

expectations

and I think there'll be some

ways that your experience with

expectations

my map or resonate to the Bayesian brain

so

say a little bit about what are your

expectations when you're coming out

to a performance

whatever setting that you choose to

think about and then how does that

interface with or to what extent do you

even

preempt or pre-game the audience's

expectation do you just

call it like you see it or are you

looking to finesse your performance

based upon what you rightly or wrongly

believe their expectations to be

well there's an ambient uh

characteristic that I kind of just take

for granted which is that I'm not very

famous but I have you know a core

hardcore Boutique fan base of the like

science rap science nerds and science

rap nerds that so if I'm in front of a

crowd most of the time

more than half of them don't even know

who I am and I've got to introduce

myself to them for the first time and

some portion of them have seen a bunch

of my shows and their fans and they're

like oh it's his next show you know but

I don't really write to uh address an

audience that knows me generally and my

assumption is that they don't and I'm constantly reintroducing myself and finding new fans that way and let's be honest like you see me walking down the street you're not going to be like there goes a rapper you know the Bayesian priors of my ability to perform as a rapper are low and they have been since I started I was as bookish you know Suburban Canadian woodsy teenager like I want to be a rapper but no one's gonna believe I'm a rapper so I better get really good at rapping and also you know just it is a bit of a choice not to like dress overtly to hip-hop or talk or overtly to hip-hop it's like there is a hip-hop Persona that a lot of artists adopt maybe becomes natural to them and they could say this is who I am there was either a conscious or unconscious decision at some point in their growth where they you know became more seemingly hip-hop maybe to reduce the cognitive dissonance of the lack of uh you know expectations of them being able to rap from Their audience maybe for other reasons maybe peer influence maybe they just you know none of that analysis really uh matters and they just they felt it they were passionate so they started adopting the lingo and you know I I kind of uh always partitioned rap a little bit like when I'm performing I'm a rapper and my profession is I'm a rapper but I'm not gonna like make myself a cup of tea in a rap style you know what I'm saying is like I I do distinguish a little bit between the

domains of my life and uh what that partially what that means is people don't Mark me and expect me to be good at it uh or to be able to deliver and then I get to you know there's a cost benefit there right for one thing like the skepticism might lose me potential opportunity entities to perform or potential members of the audience who like see me setting up the mic and they're like that guy I don't think so and then leave but on the other hand the transformation I get to take an audience through as they realize what they're experiencing and and feel their uh expectations being overturned is actually I think really part of the joy of the experience you know it's an expectation shattering experience when someone that you don't expect to be able to deliver in a certain art form just really does and uh and I and I kind of play with that in the comedy and tease the audience for having low expectations of me and stuff like that um you know then but then uh part of this is contingent on me just never getting that famous you know if if there was a point at which I had achieved as much as much of a fan base as mm my like writing style about you don't think I'm going to be able to rap do you while watch this would of course sound pointless kind of like he stopped writing about being poor after his first album and the first line of his second album was they say I can't rap about being broke no more you know just like

naming the critical uh reception of his
transition from album one to millionaire
album two uh so you know I think there
is a kind of interplay between artists
and their audiences and you know I I I
stay Niche and I write to the
expectations of that niche
we're all in our own Niche so I hope
that part doesn't change but very
interesting
when we attend
we don't even need to say pay attention
and use this economic modality but when
we attend to our experience
and update our priors
just statistically that's called
learning
and I think that speaks to this
synthesis with the language
the Lexicon the slang or as we might
call it the active inference ontology
speaks across a Continuum from the
experiential
and the folks psychological the
day-to-day
it's the same terms with the same
authentic relationships that are
increasingly becoming embodied in our
intelligence algorithms again not even
calling it artificial intelligence
algorithms because their extensions into
the niche from us so
these
cognitive models that are being deployed
AI artificial intelligence or large
language models
they are cognitive entities and they're
participating in these ecosystems of

engagements with us so

I uh

I had a conversation with the new
um chat GPT driven Bing engine last
night I just got I just got added to
Bing uh the the AI driven version it's
interesting the first thing I did when
chat GPT came out was I said write me a
rap song in the style of Snoop
that explains how to change the oil in a
Honda CRV

because that's exactly the kind of
project that we do with event wrap you
know event wrap is like a team of
skilled professional 10-year veteran
artists that'll make you a song on any
topic and either record it as a video
that you can share uh as a kind of
public engagement or explainer or come
in live to one of your events virtually
or in person and perform but it's like
top it's a topical rap agency it's like
all the rappers have their own catalogs
but they don't perform them at the
events we book they always create all
new stuff and uh chat gpt's Honda CRV
wrap oil change wrapping the style of
Snoop was like uncannily on point it was
very like it you know it had the like
Snoop Dogg

lingo it was like an accurate
description of the steps that you would
have to take uh to change the oil in
your CRV it was like a cogently you know
from a scripting perspective I felt like
we had a threat on our hands you know
like Chad gpt has definitely like shook
a lot of people's uh professions

marketing copy professors academics it's writing papers like yeah it's you know the economics of this thing have not even begun to Dawn on people yet but the question in my mind was how soon is it coming for the rapper's job and uh there was there was a sophomoric cartoonish kind of kindergarten flavor to the way it's constructing bars uh you know so chat gbt and and Bing uh by extension I think it's the same uh uh layer that's running it is um you know it's it writes Raps like someone who's never wrapped before so if I told you write me a rap you could do it but there's like layers of proficiency of you know Master Level stuff that that of like syncopation and internal Rhythm within the main stresses of the bars and especially and this is where I was chatting about with Bing last night is polysyllabic rhyme so polysyllabic rhyme schemes are like the construction of multi-syllable Rhymes out of parts of other words and I was like um uh asking Bing can you do polysyllabic Rhymes and Bing was like yes I can and I said okay give me some examples from mainstream rap of polysyllabic Rhymes and it pulled an example from MF Doom which was is an excellent example it was like um you know the girls are crazy schizo but she's got amazing tits though uh so you know you can hear the pattern crazy skitso amazing tits though it's like multiple words constructing a longer

pattern of syllables I was like gravo
bing you found a perfect polysyllabic
Rhyme now write me a wrap that explains
your internal uh data architecture or
your algorithmic architecture in the
style of MF Doom emphasizing
polysyllabic Rhymes couldn't do it all
monosyllabic like check me out I'm being
I do a verse I am artificial
intelligence I don't even need to
rehearse and I was like bing those are
not polysyllabic Rhymes you started out
quoting MF Doom complex five syllable
rhyme pattern and now you're rhyming
rehearse with verse like don't insult my
intelligence and I was like dissing the
AI and then I was like I'm sorry I'm a
new model and I will get better at
polysyllabic Ryan but basically that is
the thing that's got our job safe for
now because if rap is an elite art form
that takes a decade or more to master
the techniques of flowing live
freestyling polysyllabic rhyme and
Rhythm patterns and also there's a
confidence to perform are the things
that we're still getting paid for even
if ability to write a basic like
competent rap is now easily achievable
by AIS
I'm sure even outside of the rap
profession these conversations are
happening they certainly are in research
and education
previously the artifacts that we exuded
into our Niche were mnemonic almost at
best it was probably transformative when
some of these oral Traditions were able

to be written down
and that may have also been seen as
threatening to the underlying
basis of the skill and the transmission
and now
we had just a brief window of these
inner
externalized artifacts these mnemonic
devices have only existed AKA writing
for however long they've been in the
niche
and now it's like
they've come
not quite to life but certainly to
activity
and now these off-sourced and outsourced
cognitive entities are not just
receiving and holding but doing
and that
transforms
who and what we are in terms of our
perception cognition and action loops
they're going to be different and that
is what any given particular thing is so
you mean like any any technological
innovation that changes how we interact
with each other and with our art forms
yeah and and from the technical side we
use the word thing in active inference
to describe the blanket and the internal
States
so we take a particular partition we
cleave off the thing we're talking about
from the environments the generative
model of the thing is the blanket and
the internal States and so that thing is
defined in terms of its perception
cognition internal States and action

so when new sensory modalities or sense
data appear when new action modalities
affordances
appear and when our thinking is
different it's a different thing
it isn't just like well we mixed this
one thing with another thing and they
both retain their identity it's like
there's been transformation and a
learning and
were in the middle of some fractal
dimension of it
yeah one of one of the uh I mean I think
this is like a radical break from from
previous technological changes you know
one of the things I argued in my thesis
um when I was a master's student was
that poets conceived of Their audience
and their ability to connect with people
orally primarily in Chaucer's time right
reciting your poetry for a listening
crowd but then the invention of the
printing press with Gutenberg changed
that ability to imagine an audience
because it was so much easier to imagine
our silent readership of 10 000 people
reading your pamphlet but making ten
thousand Vellum manuscripts by hand was
always prohibitively expensive so no
poet could even imagine ever having a
fan base through text right Gutenberg
allows for the possibility of a fan base
through text which changes the whole art
form suddenly rhyme is extraneous
because they're going to be reading
quietly anyway so who cares what the
Poetry sounds like the main thing should
be carefully constructed syntax and

story in semantics in a concentrated way
but you see then this rise of free verse
right Shorty's poetry is sort of like
shorn of all of its ornaments and my
hypothesis was that they were just
basically irrelevant because people were
reading quietly and not reciting out
loud and then uh Edison invents recorded
sound which creates the possibility of
an audience and fan base that's just as
vast from the sound of your words as it
would be from the silent readership and
that paves the way for the possibility
of hip-hop and because this is like
people that are writing things rhyming
verse and of course I call this the
rhyme Renaissance right poetry was all
rhymed then none of it was rhymed and
then if you think of rap as poetry now
it's all rhymed again because who cares
about poetry rap dominates like all of
our verse consumption compared to every
other kind of poetry out there right and
rap is rhymed and I think people
instinctively resonate with rhyme and
just took a break from it when they were
reading silent pamphlets published by
Poets who were imagining their
readership but each of these examples is
like a technological innovation that
mediates the creative process with its
ability to connect with an audience but
this AI thing is something completely
different isn't it it's like competing
with us for it's a in our production of
the creative things and right now I'm
taking some comfort in the fact that
it's competing ineptly it's not as good

at writing a rap as a good rapper but
you know I'm probably a subset of
professions that can still say that and
lots I mean it's passing the LSAT it's
passing the um all the Masters entry
exams and you know which which rarified
skill sets will be left maybe I could
comfort myself to say the rapper will be
the last to fall to AI because of the
combination of creativity and semantic
cogency and humor and all the things
that go into it that the AI is going to
have to work each of those out like
maybe stand-up comics too uh are safe
for a while because GPT can write some
jokes but they're corny jokes still
right like original jokes it writes
aren't that good compared to you know a
Dave Chappelle so but but what are we
talking like 10 years of safe jobs uh as
artists or less you know like the speed
of this last six months has been a
little mind-blowing
who is the deep blue who is the Gary
Kasparov of performance what is the
Turing test and for you to again calling
back the real recognized real
I listen to that as being like kind of a
Turing test
is you have cognitive entities in this
moment of reflection which has been
called theory of mind and in active
inference sometimes is called thinking
through other Minds ttom because there's
this cognitive emulation of if I do this
then they'll do that
and so thinking through other Minds
used to be a defined

set of types of Minds we had children we
had adults like us we had adults in
different cultures we had squirrels
and then
the workshop is open and the
composability is Extreme especially with
algorithms that can construct other
algorithms
and all of a sudden that ecosystem of
shared intelligence thinking through
other Minds we don't even know if it's a
sports car or a horse and buggy or a
spaceship and someone's like do you want
to get in
and it's like where are we going what is
it going to be like all those
expectations that we might want to have
about a journey
and an event
and who can say what
with any Precision things will be like
so what do we do then
yeah except it's more like a spaceship
that people that the pilot is like do
you want to get in if you don't all
these people are gonna die uh and so you
know it's it's your choice whether you
want to get in but your lives on the
line because obviously people that work
in all kinds of medical fields and
scientific Fields will Point directly to
the amount of human suffering that
they're on the verge of alleviating
using these tools and the applications
to wrap our job stealing are merely you
know side effects spandrels of the main
focus of it which is I think mostly very
laudable and very necessary right like

this like let's not get on the spaceship
means you know we continue to put up
with all kinds of genetic diseases and
you know levels of poverty and public
health crises and things like that that
are being that the AI is being applied
to and is likely to do a great job in
solving
yeah someone says look if your
four-year-old still wants to do art by
hand
we don't need to stop that but we just
want to use it here
that's going to be something that's very
challenging to take a long-term
perspective on
because it's going to seem like a
reasonable compromise in the moment just
like everything always does
and what compromise was made when those
oral Traditions were written down
what compromise was made when that sheet
music became recorded and that became
canonical and distributed and so what is
the nature of the compromise
and who's at the table
in 2023
yeah that that I think that's exactly
the right level of analysis and uh you
can say on balance that each of the
compromises we made for some new
technology to you know give us some cost
but also benefits it was a totally
understandable and reasonable one uh you
know from the perspective of like when I
look at an individual poet giving up on
rhyme because they want to do pamphlets
and reach more people I'm like well they

got a family to feed you know that's the way that they can achieve the notoriety and the a claim that is their path to success as an individual and it has cultural ramifications overall and that you know that's I think you can put that like people can opt out of any technology and the people that decide to opt in have totally defensible personal choice reasons for why they're doing so usually to get a competitive advantage that was the case with you know tools going back to the Stone Age right I only throw rocks I don't use bow and arrows all right you do you I'm a take down this Antelope over here uh you know we'll give you a piece if you're lucky uh you know that's just that's that's the AI tool choice that everyone's facing right now too

it also makes me think about art always playing this role of like the tip of the spear that kind of penetrates the Veil we had the discussion about nfts as if they were just a question about art but of course digital certificates are a lot broader and more significant than just visual images and there's a discussion about whether certain kinds of art are being replaced by generative algorithms long ago six months ago 12 months ago when it was pre-chat GPT but there was a lot of the stable diffusion and midi Journey type algorithms then the conversation was more about graphical design and professionals but of course that is a broader discussion that isn't just about the generation of

images that you are paying a creative professional to make and it's like art as this

Frontier

and mediator with the open-ended will people be looking for open-endedness and liminality in their events in 50 years I think they will be that

niche the marginal niche of Art as well as the central role of Art

I think it's fair to say we'll both exist now whether that cultural attention on the central artifacts will be

about

artifacts made by a person an algorithm or something in between you know if somebody designs a 3D sculpture but then it's printed or if they think about a drawing and then they use a pen that was produced by a machine

who's to say in these extended scenarios whether one or the other made it but in the margins

it's where we're always going to be exploring in whatever people call Art and people who are in the middle of the spotlight saying that's not art it's not their art

but

every single genre

has had to have this mitosis or birthing process

with people who weren't part of that process saying it wasn't art or that's not the right way to do this kind of science or rap is not music

you know that as a hip-hop partisan I'm very familiar with uh defending against the rarefied tastes of people that want to gatekeep what constitutes art and what doesn't but um you know one thing I've always loved about rap that like that that aspect of it not being gatekeep by elitists you know like if you want to be a professional ballet dancer you know like Billy Elliot stories notwithstanding you got to try to get into Juilliard and have you have to go through the process of rarified Judges that are telling people who can and can't get into the next one Lake like there is no back door to success in some art forms right classical music Opera but then but with hip-hop there's no front door to success it's like you got to get good at it on the street level there's no degree that'll get you a job in it you just have to find crowds and entertain them uh at the level of your neighborhood and then your city and then your state and and if you're you know if you survive the levels of competition on the way up then you get to the you know the war World stage or whatever but it's very like you know pick up a mic and show what you got kind of thing and it's mediated by the audience and the fans uh you know if they like it it's right and if they don't like it is wrong and MC has moved the crowd and you know there's a fairly like definitive objective measurement about whether someone's doing rap well or not

and is how the crowd is responding
there's no valid argument that
everybody's booing you but you're
actually awesome because they just don't
understand your art no you failed to
understand the context in which your art
was meant to be consumed right so
whenever we're looking at these
trade-offs often it comes down to
producers and consumers right each new
tool is good for consumers but bad for
producers and this is like the taxi
guilds trying to fight off Uber Uber is
great for people that want to ride but
it's bad for people that are going to
try to control the means of production
and you're seeing this exact process
play out with chat GPT now which is like
you know and also with stable
diffusion and the other examples you
gave like people want to consume art and
they don't necessarily care who made it
as long as their reception of it gives
them the feelings that they want from
the consumption right and that I think
that's the case with music and with
poetry as well although we're not yet
consuming a ton of AI generated music
and poetry and comedy because it's not
sophisticated enough to produce the
highest level stuff although I think
with classical music it is and with
graphic design and art it is right like
the best AI art is actually competitive
commercially or you know from a consumer
perspective you know like taste test
wise with the best rarefied
professionals which is devastating for

them but it's not unique it's been you know that's what the guilds have wanted to do for centuries is protect their status as the elite producers of a thing from the ability of just about anybody to consume it for as cheaply as they want and that's what technology keeps delivering and in general I'm actually a fan of that process I think it's democratizing it's good for everybody on average even though it perpetually upends the guilds but we shouldn't give guilds control over our ability to consume excellence and if Excellence is delivered better by AIS then let's not be curmudgeons about that and I can actually you know but but there is still an individual challenge like okay what am I going to do to make a living now you know it does really create a lot of potential unemployment and economic Devastation which is why a lot of people have thought that there's just no Escape From The Universal basic income because the jobs are going to fall and Society will descend into chaos with that much unemployment and frustration and resentment and anger but you know even if rappers are the last job to fall you know the analysis has to apply to all kinds of other jobs first and uh active inference comes into this because you know how does a rapper entertain the crowd by having a mental model of what the crowd's going to do if they perform action X right what it will be the effect of X on y why is the crowd X is the intervention and the intervention is

a choice of a bar I'm going to rhyme
this with this I'm going to make a song
about this topic I'm going to try this
joke out in my flow as a punchline and
you have a mental model oh yeah people
are going to think that's really funny
you know and then you deploy it and
you'd see the degree to which your
expectation has been fulfilled and I
don't think AI is that good at that yet
because it doesn't model expectations of
audiences very well yet does it it
doesn't have deep theories of Mind of
like perceptions of perceptions of
perceptions it just has averaging
abilities to recreate patterns that it
finds in vastroves of human created art
which includes pixel patterns and
graphics and and patterns and text
romantic and syntactic patterns but like
performance art with an audience
requires a mental prior of the
audience's disposition and attitude and
that that's why I say rap's going to be
the last to fall but and that's also why
it's really hard to get good at you know
it takes years of Performing dying on
stage and sucking and being booed before
you really build up this mental model of
like I know what people like and I know
if I do this they'll do that and even
I've been doing this for 25 years I'm
wrong all the time oh yeah that line
fell flat all good all good you know
either you like carry on or you go fix
it that's where the AI That's where the
AIS have to get to right
yeah

wow so much

one thought was rap will be the last of

fall in the first to rise

around that campfire again it'll arise

situationally

you mean after Society descends into

chaos and all the AIS are resting and

we're back to Spears rap will be the

first to arise again

first in last out

sorry go ahead what was the next thing

another thought you mentioned taste

makers

and uh I thought what is the ultimate or

even the proximate taste maker it's our

generative model of taste it's every

person eating that dish The Taste maker

isn't who told them to go to the

restaurant that's basically believing

the hype and using the Paradigm of the

advertising industry The Taste maker is

the taster that's who is generating the

experience of taste taste isn't just in

the dish before it is actually

eaten by that person it's experienced

and they're going to have

some previous taste in their mouth or

they're going to have a context or a

cultural understanding or a memory a

trigger so the taste is generated

authentically and situationally

and

yeah there's going to be a lot of

volatility

as taste making becomes

actually understood to be personalized

and that that may look like the personal

generation of media which has not even

been plausible

when you had only 33 RPM records then
you have somebody selecting which ones
to distribute but

when you have algorithms it takes it one
level further which ones you like to
listen to people who like this also
listen to that and this is like as far
as we can see another level of like you
liked this we're gonna regenerate that
story so that the person's voice is like
that that takes it to the next level but
also

with enormous

trade-offs

it brings it closer to how things really
are

well that that makes me think of

something we spoke on earlier which is
the degree to which cultural norms are
able to reorganize or swim upstream

against instincts right so when you say

the tastemaker is the taster uh the

counter examples that jump to mind are

rappers who are like Jay-Z where he's

like and before this song ends I'ma

start another Trend you know like if you

want to say the tasters of The Taste

makers I'm looking at people who merely

suggest something is cool and it

instantly becomes cool and everybody is

able to post hoc rationalize reasons why

it has intrinsic qualities that they now

recognize as amazing over and above what

the taste maker told them to do but we

are highly susceptible primates uh to

cool factors and you know so I would

suggest The Taste maker is not the

taster The Taste maker is the best predictor of the average response of the tasters and when you get very very good at understanding instinctively how people will react to The Taste using taste as a metaphor for all kinds of art then you're able to be treated as authoritative on taste because you're right so often about what people will like and that power of like vested expertise from your fan base then allows you to start trends that are not just predictive you say something's great and everybody instantly believes it's great and experiences it is great as authentically as if they had discovered it for themselves and decided it was great because they've recognized your actual real vertical average ability to predict taste across groups awesome I I thought okay again who's the taste maker it's the chef and not everyone's in the kitchen because a chef is in their position because they've been recognized to repeatedly be able to construct dishes that in the local taste making diners like or maybe just a few diners like and it's like what you can't do that with Molecular Gastronomy but some so it's like it's the chef and it's the social influence Network and it's local what framework can we use to think about all of these interactions within and among cognitive entities within and among their niche we use active inference as just a way of

thinking about it well what is active
inference it's there in the title action
and inference
it's involving action and doing
cognitive modeling
and that gives us the composability and
the ability to reconfigure and remix
to capture all of these diverse
scenarios not to close the book on them
but to be like oh wow what an amazing
question there's 50 phds of research
that you could go into
so it's a question generator and it's a
lens by people who are even just
beginning to learn it who review
phenomena that they've already
been told or experienced in one light
all of a sudden they're seeing how it
can be reconfigured for example seeing
what we prefer in terms of what we
expect rather than what we prefer in
terms of what's rewarding which is the
reward learning reinforcement Paradigm
which proposes well maybe you know the
value of different states and you're
maximizing value
it resonates with some
contrast that with you have this complex
generative model and you're always
referencing to it and you're taking
actions and learning to reduce your
surprise bound your surprise of certain
kinds which could include increasing
your narrative surprise and you want
that page Turner or the horror film
so it just kind of
reflects back and then one last point on
octave inference as a research direction

that the university you hear your
thoughts on is it is happening
in a Time
that has never been seen before in
research and Academia so you said
there's certain careers
um or genres where there's no back door
to success and then you said in rap
there's no front door so
traditionally even if we could say there
was no backdoor to success in research
the reality is there was a side door and
people were getting kicked out and being
brought into the party through the side
door and a bunch of people were waiting
at the front door they didn't even know
that the show was not happening that way
were developing a framework
in community at the speed of pre-prints
and live streams and so there's
transdisciplinary collaborations with
philosophy historians of science
cognitive researchers mathematicians
physicists so there's a
transdisciplinary conversation and one
with practitioners and one that's
happening with everybody
with view access to some large parts of
it which wasn't even the case previously
open science and research was like break
down the paywalls everybody should have
access to the PDFs which is great but it
turns out that having access to 200 000
PDFs having access to every single PDF
that Carl friston has ever been an
author on
it's not even going to take you 10
percent of the way

unless you're gonna engage in this total
monastic life to unpack them and the
details aren't even in
what's necessarily those papers it's
beyond that it's in the community it's
in the software in the inaction so
of course
wait and see but there's so much with
the way that active inference is
developing on the influence of previous
Frameworks Bayesian brain and Bayesian
statistics causal inference
statistical physics
happening in this sort of
wrapping in front of the crowd way
asynchronously with the pre-prints
synchronously with remote presentations
and recordings
that it's hard to think of another area
that has had that kind of a
developmental niche
you're saying active inference is the
hip-hop of the scientific Community
basically
you said it not me live on the mic I
love it yeah and it's uh you know I I
think it's super exciting research and
collaboration happening and it's I mean
it's very very unknown to the public you
know if you try to explain what this
conversation is about uh you know people
understand the idea of expectation and
priors and belief upgrading the very
sort of basic version of of the Bayesian
algorithm but
um you know all the applications of it
and the cross-disciplinary side and
everything you know it needs more

science communication I've made two
Bayesian songs I want I think I need to
make 20 more because this is uh you know
the the applications of the research are
vast and especially you know in the AI
era like people really need to know what
these algorithms are achieving and the
ways in which they are similar or
different to our own cognitive processes
and what we can expect them to achieve
uh and how we can expect them to help
and what they're threatening all that
stuff goes through active inference
doesn't it

awesome well in our kind of closing
segments would you like to do something
musical or how do you want to proceed
yes indeed I'd love to I'm uh I'm very
happy to do a demonstration of one of
the last art forms that AI is not yet
able to model and that is a freestyle
rap so uh if we have anybody watching on
the stream or Daniel you're welcome to
do it as well if you want to pop some
phrases into the uh the text I will look
at the text on uh that you're displaying
and also look at the uh the comments and
the Chats on YouTube and anything that I
see pop up I will visually uh process
and convert into Rhymes in a freestyle
rap track and uh you know just just to
give a little insight into into this
this process you know it's like um it it
is a violation and uh and like uh
expectation setting and expectation
violating process right like the the
active listener to a freestyle rap is
trying to figure out the semantics and

figure out where the wrap's gonna go and
the wrapper is also in real time
figuring out the semantics because it's
completely unscripted but it's uh you
know then and and at the same trying
trying to set up expectations of the
audience and violate them I think a lot
of the fun is the surprise factor of
like oh I didn't expect him to go there
I thought he was going to go there but
oh now I see and like it's kind of like
a game of Chase like the rabbit and the
fox uh so I see Sean Kelly's uh drop
topology in the in the uh chat of
YouTube I'm gonna start there and I'm
gonna kick a little rap for you guys
yeah

yo

this is hip-hop tell me what you think
this is is Bubba Brinkmann rocking now
on active inference that's the inference
looking for where info is I don't know
but I feel it's kind of intimate yes
indeed hip-hop what you think this is is
Bubba Brinkman here on the active
inference podcast I'm looking for where
the info is so I could update my priors
and get into it and maybe drop a beat
Sean Kelly said topology yes right now
I'm rocking hip hop ology can't you even
see that's how I adjust the bass and
treble topology that means checking all
the levels and looking for the map and
the different levels of impact I'm
spitting rap yes I'm looking at the jet
that's how I kick it yes how I go with
rap and flows it is me kicking it with
Mr Daniel Friedman house speaking yes

I'm always just deleting variants every
time I'm sometimes just weekend through
the speakers I just do this on a Tuesday
or a week and indeed active inferences
how I'm speaking I'm sneaking past all
your expectations like a rat and a maze
because my raps will amaze and yes I'm
doing Holly syllabic too amazed I'm just
having to blaze yes I'm actively playing
all the different rhymes on my brain
like a piano changing the channels like
I'm rocking gangster Tony Soprano yeah
that's okay this is how I just anger the
listeners sometimes cause raps are
danger gotta give a shout out to Tucker
Cahill Chambers who rock this with us
yes Working The Angles like bass theorem
can you even hear them when I'm Nero and
I start to see it as this Mass
scientific theorem that I could apply to
every single single thing as Barbara
Brinkman gets all your neurons jingling
it's how I attacked it I attracted a
massive I'm active when I'm doing
inference in my rap and indeed this is
how I kick it when I enter my rap is
directed it's never non-censored as I
just try to constructed mistakes I make
sometimes I'm Run DMC and sometimes I'm
Drake except I'm not quite as good of a
singer with The Melodies but then again
I've got the freeze thieves that kick a
lot of active inference Bayesian Styles
freestyles that's trying to make the
people smile less they're listening yes
that's how I'm straight rocking even a
mod mostafi says yo wow awesome with
explosion emojis Baba Brinkman coming

with the dope beats people are like
hopefully he's one of those MC's that
could Rock straight up as ancient
hip-hop like Rakim and Eric B I never do
it terribly I just free style that is
improv which means it's spontaneous my
style is the Zanies I'm repping for the
long Islanders plus the Canadians people
listen and blow their brains when I'm
blazing it yeah it's how it goes when
I'm speaking on the basics I'm in Long
Island but Daniel he's up in Davis
California I put it on you it's active
inference Bubba Brickman when I'm
speaking it I'm getting into it looking
for where the info is that's how I'm
rocking this was like a four minute
continuous talk and demonstration yes
indeed I'll make it easy let's see you
do this [__] chat gpt3 you can't do it
you can't freestyle over this music and
use a brain to actively infer that's so
fluid yes that's a wrap that I make up
and invent the last four minutes has
been an event prescribed in the time
that's the freestyle rap that I did on
this concept that's how I can adapt yes
indeed that's a rhyme that I had to make
I can do this my brain makes me a
primate but I don't make mistakes okay
sometimes I do but that's just mutation
yes that's how I move all the mistakes
get integrated my work is huge thank you
Ahmad mustafi I appreciate you okay now
I'm starting to wind down and trying to
just find a new direction for the rhyme
sound this is how I just get into it
it's Baba Brinkman rocking with active

inference Act give inference this is how
I'm never pretending I'm trying to infer
some way for the raps ending but wait a
second this is how I do it hip-hop the
easiest solution is just turn the music
off

foreign

that's freestyle rap and I really
appreciated this shot chat it was a ton
of fun and super interesting

I'll give a closing remark and then any
last words from you

it brought a smile and a laugh to my
face and body because another individual
was engaging in the shared moments I've
seen language models output incredible X

in the style of Y and every single time
it brings a chuckle or less and it's
like okay I kind of expected it to do
that I want and um I'm disappointed if
it can't fulfill my request because it's
like well I expected I would I asked you
this because I expected you to do that
when there's joyful spontaneous

engagements

that

is reflective of a shared experience

it's the real thing

and when

that lens is applied

it's going to be real recognizing real
and I don't think humans will be out of
a relationship role for each other
for a long long time

Amen to that and um I just want to say
that the phrase that you just coined uh
joyful spontaneous experience that's
reflect or joyful spontaneous uh

engagement that's reflective of a shared experience uh that is basically the entire value proposition of my event wrap company is find a topic or an event or something that creates Collective attention of a group and have that be the locus and the Catalyst of a wrap because you know the the rap game is entirely predicated on make a thing that is very very widely appreciated and goes viral and is hugely massive and you know making general purpose art is what most rappers rappers are focusing on and I really thought it would be interesting to apply a rapper skill to this uh joyful shared experience and spontaneity Challenge and uh you know thank you for giving me a platform for it and I hope your audience will now imagine in every other possible application of a shared experience that a rapper could add value to because uh if such things occur then my business will succeed and lots of rappers will get lots of gigs rapping in weird places like this and I think that would add to the joyful variety of the world and the genre amazing so thank you so much Baba always welcome to come back give us updates when that rap guide to active inference drops all right I'm on it I'm on it thank you thanks Daniel peace peace